STD 1: Measurement (Std 1)

M5 Scale Drawings (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 85.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-M5 Scale Drawings* has been a significant contributor to the first two Std1 exams, accounting for 9% and 7.5% of the 2020 and 2019 Std1 papers respectively.
- *MS-M5 Scale Drawings* is comprised of content which can be found in *MS-M7 Rates and Ratios*, a Std2 topic that covers ratios, similarity and the associated use of scale in buildings and maps.

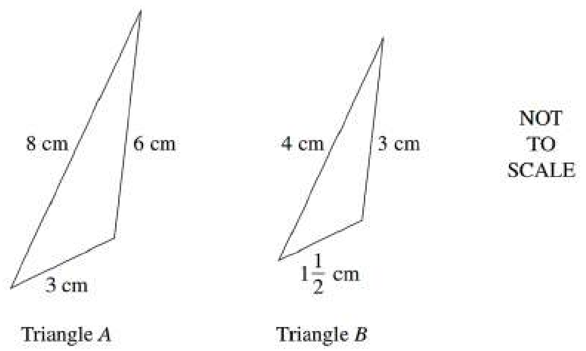
ANALYSIS - What to Expect and Common pitfalls

- Floor plans have been tested in both 2020 and 2019 Std1 exams and represent an important revision area.
- Maps and the associated scale calculations have not been examined in Std1 exams to date although numerous questions in the database cover this topic area.
- *Similarity* has also been tested in both 2020 and 2019 Std1 exams, producing red flag mean marks of 11% and 29%. A revision focus is well warranted here.
- Applied "ratios" were examined in *2019 Std1 29*, and proved challenging. Numerous questions covering 2-part and 3-part ratios are included in the database (we note a 3-part ratio was examined in the 2019 paper).

Questions

1. Measurement, STD2 M7 2007 HSC 4 MC

What scale factor has been used to transform Triangle A to Triangle B?



(A) $\frac{1}{2}$

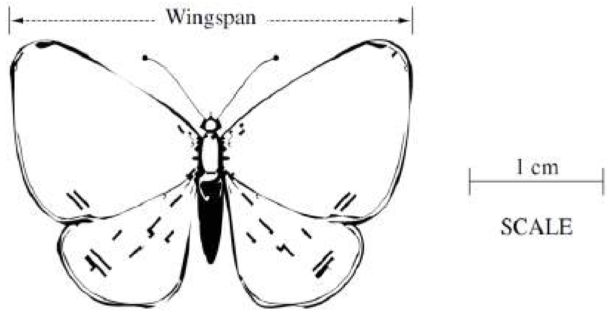
(B) $\frac{3}{4}$

(C) 2

(D) 3

2. Measurement, STD2 M7 2005 HSC 4 MC

The diagram is a scale drawing of a butterfly.



What is the actual wingspan of the butterfly?

(A) 2.5 cm

(B) 3 cm

(C) 15 cm

(D) 8.75 cm

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3. Measurement, STD2 M7 SM-Bank 3 MC

There are 8 male chimpanzees in a community of 24 chimpanzees.

What is the ratio of males to females in the community?

A. 1 : 3

B. 1 : 2

C. 3 : 1

D. 2 : 1

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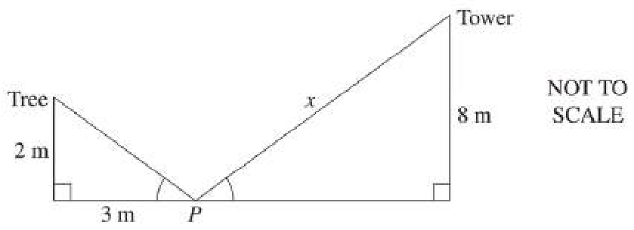
4. Measurement, STD2 M7 2008 HSC 20 MC

A point P lies between a tree, 2 metres high, and a tower, 8 metres high. P is 3 metres away from the base of the tree.

From P, the angles of elevation to the top of the tree and to the top of the tower are equal.



What is the distance, x, from P to the top of the tower?

(A) 9 m

(B) 9.61 m

(C) 12.04 m

(D) 14.42 m

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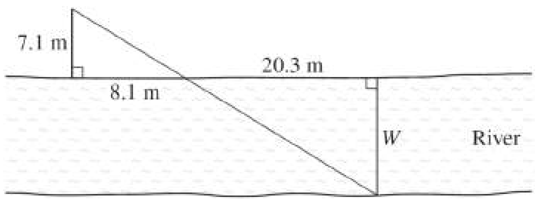
5. Measurement, STD2 M7 2016 HSC 16 MC

The width (W) of a river can be calculated using two similar triangles, as shown in the diagram.



What is the approximate width of the river?

(A) 17.8 m

(B) 19.3 m

(C) 23.2 m

(D) 24.9 m

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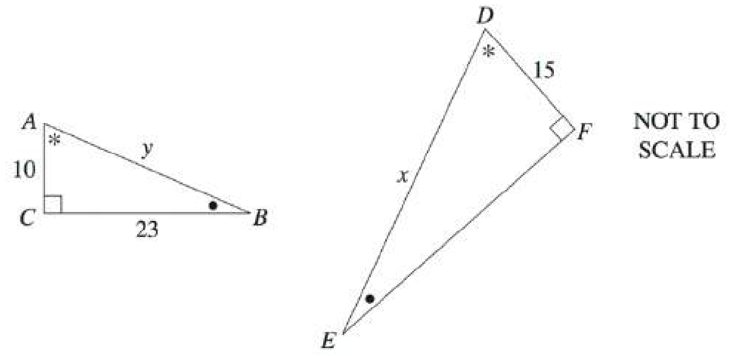
6. Measurement, STD2 M1 2013 HSC 17 MC

Triangles ABC and DEF are similar.



Which expression could be used to find the value of x?

(A) $y \times \dfrac{10}{15}$

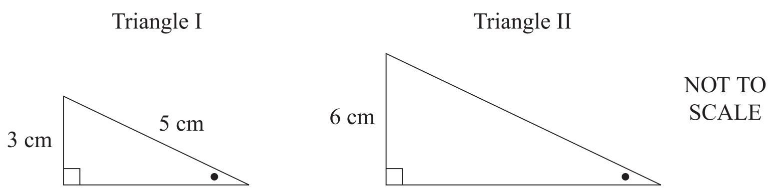
(B) $y \times \dfrac{10}{23}$

(C) $y \times \dfrac{15}{10}$

(D) $y \times \dfrac{23}{15}$

7. Measurement, STD1 M5 2019 HSC 10 MC

Triangle I and Triangle II are similar. Pairs of equal angles are shown.



What is the area of Triangle II?

A. 18 cm²

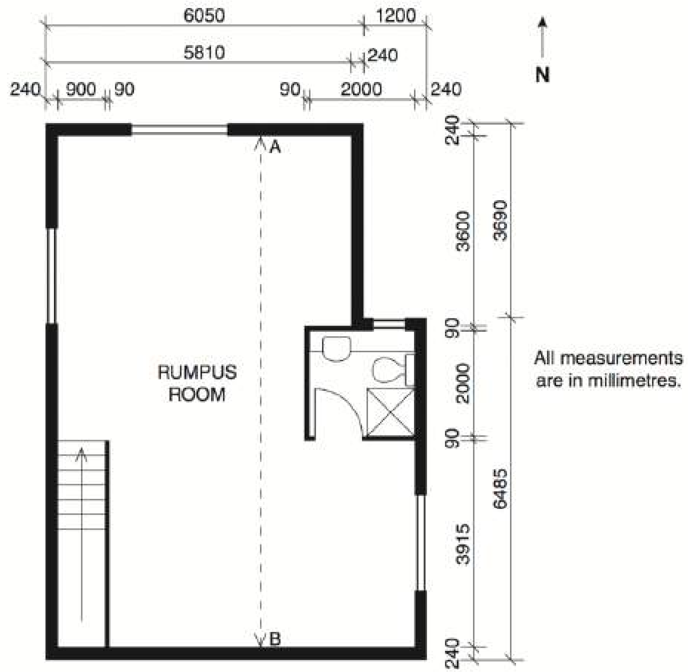
B. 24 cm²

C. 30 cm²

D. 48 cm²

8. Measurement, STD2 M7 2011 HSC 24a

Part of the floor plan of a house is shown. The plan is drawn to scale.



i. What is the width of the stairwell, in millimetres? ***(1 mark)***

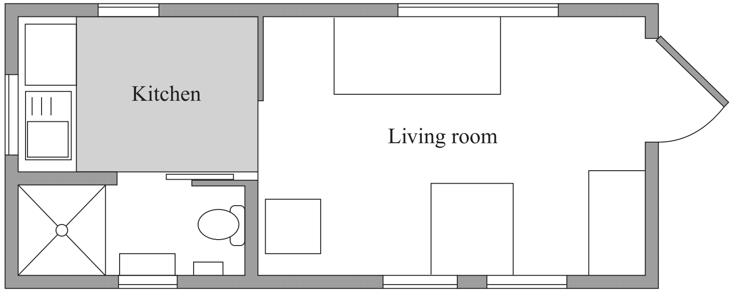
ii. What are the internal dimensions of the bathroom, in millimetres? ***(1 mark)***

iii. What is the length AB, the internal length of the rumpus room, in millimetres? ***(1 mark)***

9. Measurement, STD1 M5 2019 HSC 20

The plan of the lower level of a small house is shown.

Scale ⊢——⊣ = 0.5 m



a. How many windows are shown on the plan? ***(1 mark)***

b. What is the actual perimeter, in metres, of the shaded part of the kitchen floor? ***(2 marks)***

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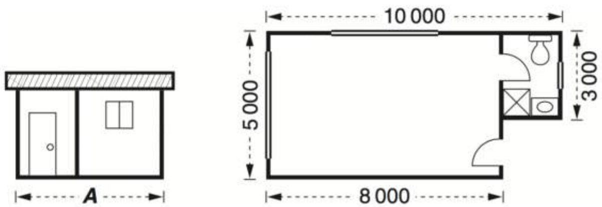
10. Measurement, STD2 M7 2010 HSC 23b

The elevation and floor plan of a building are shown.



All measurements are in millimetres.

Calculate the area of the floor of this building in square metres. ***(2 marks)***

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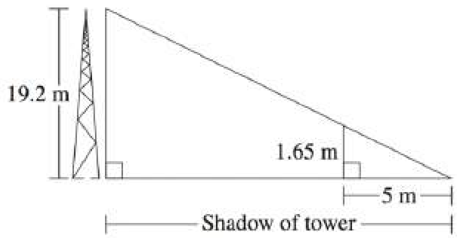
11. Measurement, STD2 M7 2015 HSC 27a

At a particular time during the day, a tower of height 19.2 metres casts a shadow. At the same time, a person who is 1.65 metres tall casts a shadow 5 metres long.



NOT TO SCALE

What is the length of the shadow cast by the tower at that time? ***(2 marks)***

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12. Measurement, STD2 M7 SM-Bank 6

In a raffle, the total prize money is shared among the first two tickets drawn in the ratio 6 : 4.

The prize for the second ticket drawn is \$360.

What is the total prize money? ***(2 marks)***

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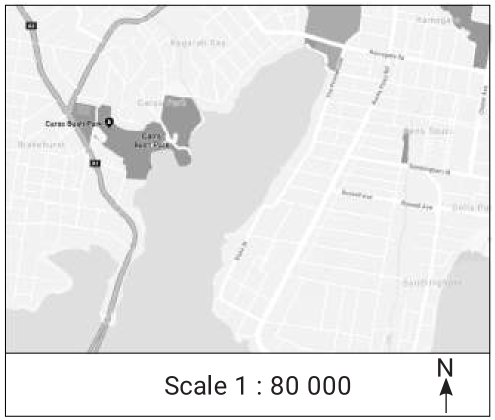
13. Measurement, STD2 M7 SM-Bank 7

Part of a map is shown.



The distance between two points on the above map (not shown) is 3.4 kilometres. How far apart on the map would the two points be, in centimetres? ***(2 marks)***

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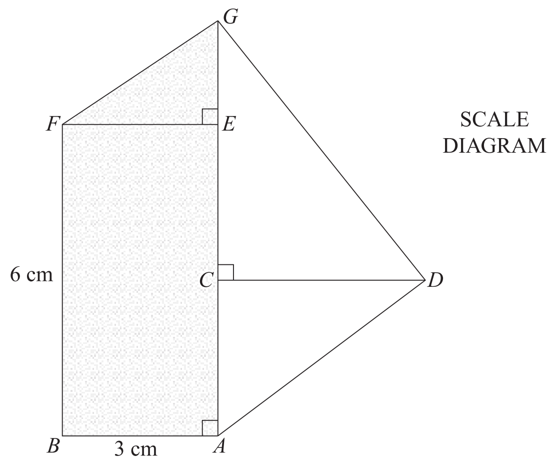
14. Measurement, STD2 M7 SM-Bank 13

A scalene triangle has angles in the ratio 9 : 2 : 4.

In degrees, what is the size of its largest angle? ***(2 marks)***

15. Measurement, STD2 M7 2018 HSC 26g

A field diagram of a block of land has been drawn to scale. The shaded region $ABFG$ is covered in grass.



The actual length of AG is 24 m.

i. If the length of AG on the field diagram is 8 cm, what is the scale of the diagram? ***(1 mark)***

ii. How much fertiliser would be needed to fertilise the grassed area $ABFG$ at the rate of 26.5 g /m^2? ***(3 marks)***

16. Measurement, STD2 M7 2012 HSC 27c

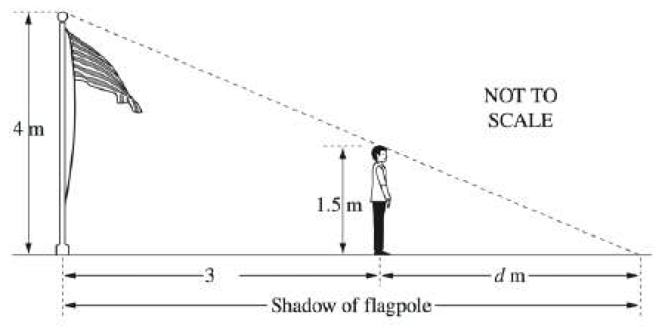
A map has a scale of 1 : 500 000.

i. Two mountain peaks are 2 cm apart on the map.
What is the actual distance between the two mountain peaks, in kilometres? ***(1 mark)***

ii. Two cities are 75 km apart. How far apart are the two cities on the map, in centimetres? ***(1 mark)***

17. Measurement, STD2 M7 2012 HSC 28c

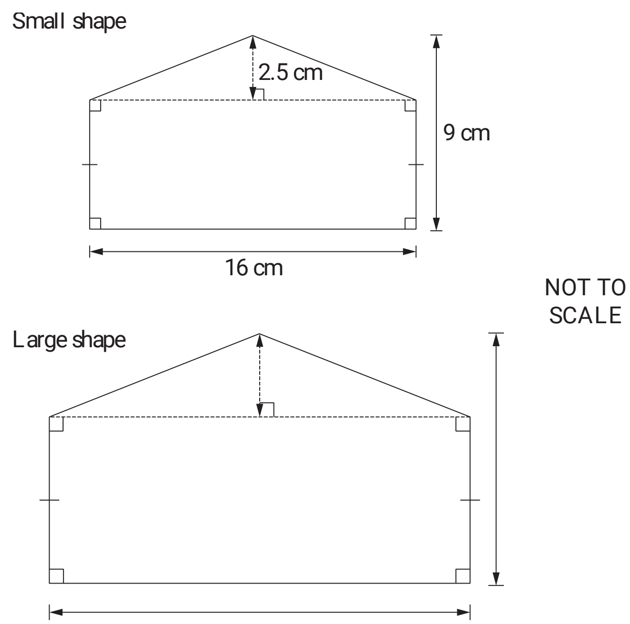
Jacques and a flagpole both cast shadows on the ground. The difference between the lengths of their shadows is 3 metres.



What is the value of d, the length of Jacques' shadow? ***(3 marks)***

18. Measurement, STD1 M5 2021 HSC 26

The diagrams show two similar shapes. The dimensions of the small shape are enlarged by a scale factor of 1.5 to produce the large shape.



Calculate the area of the large shape. ***(3 marks)***

19. Measurement, STD1 M5 2019 HSC 29

Concrete is made by mixing cement, sand and aggregate. Different types of concrete are produced by changing the ratio of the mix of these materials.The table shows the ratio of the materials for different types of concrete and examples of their common use.

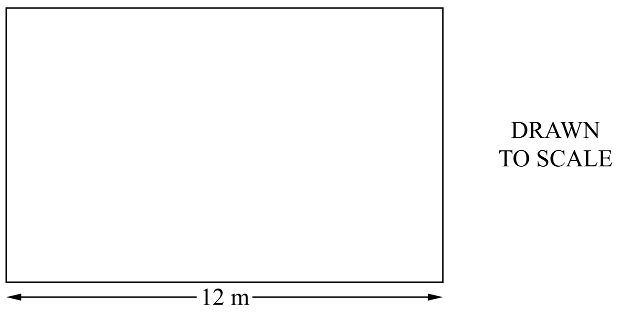
Ratio of cement : sand : aggregate	*Common use*
1 : 2 : 2	Foundation for fence posts
1 : 3 : 6	Footpath, patio slab
1 : 1 : 2	House slab

The amount of concrete required for a patio slab is 3.5 cubic metres.

How many cubic metres of sand will be needed? ***(2 marks)***

20. Measurement, STD1 M5 2020 HSC 23

The diagram shows a scale drawing of the floor of a room.

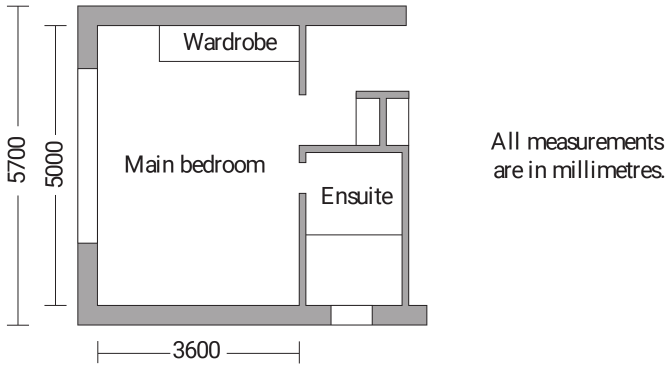


Carpet is to be laid to cover the entire floor. The cost of the carpet is $100 per square metre.

Find the total cost of the carpet required. ***(3 marks)***

21. Measurement, STD1 M5 2021 HSC 24

A building plan of part of a house is shown.



a. What is the internal length and the internal width of the main bedroom? Give your answer in metres. ***(1 mark)***

b. Rod wants to lay carpet in the main bedroom. The main bedroom has a wardrobe with a base area of 1.6 m². The area under the wardrobe does NOT need to be carpeted.

Carpet costs $40 per square metre. Rod buys the smallest whole number of square metres of carpet necessary. Find the cost of the carpet purchased for the bedroom. ***(3 marks)***

22. Measurement, STD2 M7 2015 HSC 29c

The image shows a rectangular farm shed with a flat roof.

The real width of the shed indicated by the dotted line was measured using an online ruler tool, and found to be approximately 12 metres.

i. By measurement and calculation, show that the area of the roof of the shed is approximately 216 m^2. ***(2 marks)***

ii. All the rain that falls onto this roof is diverted into a cylindrical water tank which has a diameter of 3.6 m. During a storm, 5 mm of rain falls onto the roof.

Calculate the increase in the depth of water in the tank due to the rain that falls onto the roof during the storm. ***(3 marks)***

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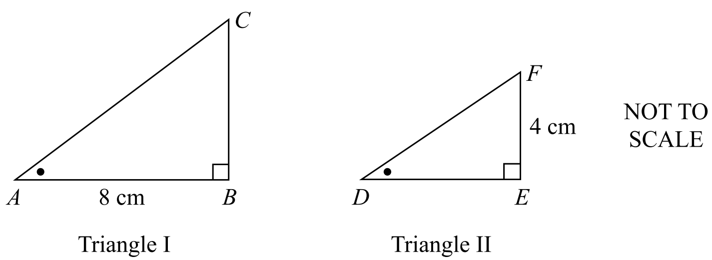
23. Measurement, STD1 M5 2020 HSC 28

Two similar right-angled triangles are shown.



The length of side AB is 8 cm and the length of side EF is 4 cm.

The area of triangle ABC is 20 cm^2.

Calculate the length in centimetres of side DF in Triangle II, correct to two decimal places. ***(4 marks)***

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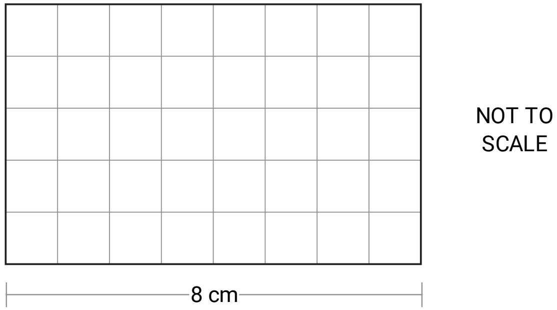
24. Measurement, STD1 M7 2021 HSC 21

A rectangular sportsground has been drawn to scale on a 1-cm grid as shown. The scale used is $1:3000$.



Kerry took 12 minutes to walk around the perimeter of this sportsground.

What was Kerry's average speed in kilometres per hour?

Worked Solutions

1. Measurement, STD2 M7 2007 HSC 4 MC

Take two corresponding sides

In ΔA 3 cm

In ΔB 1 ½ cm

$\therefore$ Scale factor converting ΔA to $\Delta B =$ ½

$\Rightarrow\ A$

2. Measurement, STD2 M7 2005 HSC 4 MC

Wingspan is 3 times the scale distance that equals 1 cm.

$$\begin{aligned}\therefore\ \text{Wingspan} &= 3 \times 1\\ &= 3\text{ cm}\end{aligned}$$

$\Rightarrow\ B$

3. Measurement, STD2 M7 SM-Bank 3 MC

Males : Females

$$\begin{aligned}&8:(24-8)\\ &8:16\\ &1:2\end{aligned}$$

$\Rightarrow$ B

4. Measurement, STD2 M7 2008 HSC 20 MC

Triangles are similar (equiangular)

In smaller triangle:

$$\begin{aligned}h^2 &= 2^2 + 3^2\\ &= 13\\ h &= \sqrt{13}\end{aligned}$$

$$\begin{aligned}\frac{x}{8} &= \frac{\sqrt{13}}{2} \quad \text{(sides of similar } \Delta\text{s same ratio)}\\ x &= \frac{8\sqrt{13}}{2}\\ &= 14.422...\end{aligned}$$

$\Rightarrow\ D$

5. Measurement, STD2 M7 2016 HSC 16 MC

Using similar ratios,

$$\begin{aligned}\frac{W}{7.1} &= \frac{20.3}{8.1}\\ \therefore W &= \frac{20.3 \times 7.1}{8.1}\\ &= 17.79\ldots\end{aligned}$$

$\Rightarrow\ A$

6. Measurement, STD2 M1 2013 HSC 17 MC

♦ *Mean mark 38%*

We know $\Delta ABC\ |||\ \Delta DEF$

$$\begin{aligned}\therefore\ \frac{AB}{AC} &= \frac{y}{10} = \frac{DE}{DF} = \frac{x}{15}\\ \frac{x}{15} &= \frac{y}{10}\\ x &= y \times \frac{15}{10}\end{aligned}$$

$\Rightarrow\ C$

7. Measurement, STD1 M5 2019 HSC 10 MC

In Triangle I, using Pythagoras:

$$\begin{aligned}\text{Base (TI)} &= \sqrt{5^2 - 3^2}\\ &= \sqrt{16}\\ &= 4\end{aligned}$$

Triangle I ||| Triangle II (given)

$\Rightarrow$ corresponding sides are in the same ratio

♦♦ *Mean mark 29%.*

$$\frac{\text{Height (TII)}}{\text{Height (TI)}} = \frac{\text{Base (TII)}}{\text{Base (TI)}}$$

$$\frac{6}{3} = \frac{\text{Base (TII)}}{4}$$

$$\begin{aligned}\text{Base (TII)} &= \frac{4 \times 6}{3}\\ &= 8\end{aligned}$$

$$\begin{aligned}\therefore \text{Area (TII)} &= \frac{1}{2} \times 8 \times 6\\ &= 24 \text{ cm}^2\end{aligned}$$

$\Rightarrow B$

8. Measurement, STD2 M7 2011 HSC 24a

i. 900 mm

ii. 2000 mm $\times$ 2000 mm

iii. Length of Rumpus Room $= AB$

$$\begin{aligned}AB &= 3600 + 90 + 2000 + 90 + 3915\\ &= 9695 \text{ mm}\end{aligned}$$

9. Measurement, STD1 M5 2019 HSC 20

a. 6 windows

b.
$$\begin{aligned}\text{Plan Perimeter} &= 2 \times 3 \text{ units} + 2 \times 3.5 \text{ units}\\ &= 13 \text{ units}\end{aligned}$$

$$\begin{aligned}\therefore \text{ Actual perimeter} &= 13 \times 0.5\\ &= 6.5 \text{ metres}\end{aligned}$$

10. Measurement, STD2 M7 2010 HSC 23b

MARKER'S COMMENT: Less errors were made by students who converted to metres *FIRST*, before calculating the area.

$$\begin{aligned}\text{Floor Area} &= \text{Area 1} + \text{Area 2}\\ &= (8 \times 5) + (2 \times 3) \quad (1 \text{ m} = 1000 \text{ mm})\\ &= 46 \text{ m}^2\end{aligned}$$

11. Measurement, STD2 M7 2015 HSC 27a

Both triangles have right angles and a common angle to the ground.

$\therefore$ Triangles are similar (equiangular)

Let $x =$ length of tower shadow

$$\frac{x}{19.2} = \frac{5}{1.65} \quad \text{(corresponding sides of similar triangles)}$$

$$\begin{aligned}x &= \frac{5 \times 19.2}{1.65}\\ &= 58.1818\ldots\\ &= 58 \text{ m} \ \text{(nearest m)}\end{aligned}$$

12. Measurement, STD2 M7 SM-Bank 6

$$\frac{4}{10} \times \text{Total prize money} = \$360$$

$$\Rightarrow \frac{1}{10} \times \text{Total prize money} = \$90$$

$$\therefore \text{Total prize money} = 90 \times 10$$
$$= \$900$$

13. Measurement, STD2 M7 SM-Bank 7

Actual distance on map

$$= \frac{\text{real distance}}{\text{scale}}$$
$$= \frac{3.4\text{ km}}{80\ 000}$$
$$= \frac{3400\text{ m}}{80\ 000}$$
$$= 0.0425\text{ m}$$
$$= 4.25\text{ cm}$$

14. Measurement, STD2 M7 SM-Bank 13

$$\text{Smallest angle} = \frac{\text{largest part}}{\text{total parts}} \times 180$$
$$= \frac{9}{9+2+4} \times 180$$
$$= 108^{\circ}$$

15. Measurement, STD2 M7 2018 HSC 26g

♦♦ *Mean mark 32%.*

i. Scale

$$8\text{ cm} : 24\text{ m}$$
$$1\text{ cm} : 3\text{ m}$$
$$1 : 300$$

ii. Area of rectangle $ABFE$

$$= 6\text{ cm} \times 3\text{ cm}$$
$$= 18\text{ m} \times 9\text{ m}$$
$$= 162\text{ m}^2$$

Area of ΔEFG

$$= \frac{1}{2} \times 3\text{ cm} \times 2\text{ cm}$$
$$= \frac{1}{2} \times 9 \times 6$$
$$= 27\text{ m}^2$$

$$\therefore \text{Fertiliser needed} = (162 + 27) \times 26.5$$
$$= 5008.5\text{ grams}$$

16. Measurement, STD2 M7 2012 HSC 27c

♦ *Mean mark 37%*
MARKER'S COMMENT: Better responses realised that 1 unit on the map represented 1 unit x 500,000 in real life.

i. Actual distance (2 cm)

$$= 2 \times 500\ 000$$
$$= 1\ 000\ 000\text{ cm}$$
$$= 10\ 000\text{ m}$$
$$= 10\text{ km}$$

$\therefore$ The 2 mountain peaks are 10 km apart.

♦ *Mean mark 44%*

ii. Cities are 75 km apart.

From part (i), we know 2 cm = 10 km

$$\Rightarrow 1\text{ cm} = 5\text{ km}$$

$$\Rightarrow \text{On the map, } 75\text{ km} = \frac{75}{5} = 15\text{ cm}$$

$\therefore$ Distance on the map is 15 cm.

17. Measurement, STD2 M7 2012 HSC 28c

♦♦ *Mean mark 24%*

Both triangles have right-angles with a common (ground) angle.

$\therefore$ Triangles are similar (equiangular)

Since corresponding sides are in the same ratio

$$\frac{d}{1.5} = \frac{d+3}{4}$$
$$4d = 1.5(d+3)$$
$$8d = 3(d+3)$$
$$= 3d + 9$$
$$5d = 9$$
$$\therefore d = \frac{9}{5}$$
$$= 1.8 \text{ m}$$

18. Measurement, STD1 M5 2021 HSC 26

♦♦ *Mean mark 32%.*

Dimension of larger shape:

Width $= 16 \times 1.5 = 24$ cm

Height $= 9 \times 1.5 = 13.5$ cm

Triangle height $= 2.5 \times 1.5 = 3.75$ cm

$$\therefore \text{ Area} = (24 \times 9.75) + \frac{1}{2} \times 24 \times 3.75$$
$$= 279 \text{ cm}^2$$

19. Measurement, STD1 M5 2019 HSC 29

♦ *Mean mark 34%.*

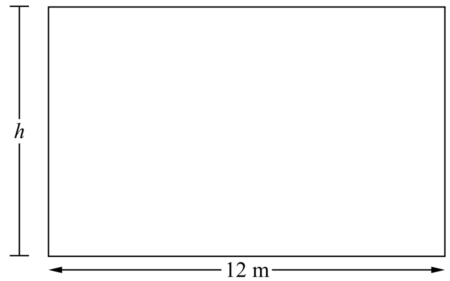
Patio ratio,

Cement : Sand : Aggregate $= 1:3:6$

Concrete required $= 3.5$ m^3

$$\therefore \text{ Sand required} = \frac{\text{sand parts}}{\text{total parts}} \times 3.5$$
$$= \frac{3}{1+3+6} \times 3.5$$
$$= 1.05 \text{ m}^3$$

20. Measurement, STD1 M5 2020 HSC 23



DRAWN TO SCALE

Ratio of sides $=$ base : height

$= 8:5$

Find actual h:

$$\frac{h}{12} = \frac{5}{8}$$
$$h = \frac{5 \times 12}{8} = 7.5$$

COMMENT: Note the image dimensions on the actual HSC paper were exactly 8 cm × 5 cm, making this ratio easy to calculate.

Area (actual) $= 12 \times 7.5 = 90$ m^2

♦ *Mean mark 41%.*

$$\therefore \text{ Cost} = 90 \times 100$$
$$= \$9000$$

21. Measurement, STD1 M5 2021 HSC 24

a. $\text{Internal length} = 5000\text{ mm} = 5\text{ metres}$

$\text{Internal width} = 3600\text{ mm} = 3.6\text{ metres}$

♦ *Mean mark part (a) 46%.*

b. $\text{Carpet area} = (5 \times 3.6) - 1.6$

$= 16.4\text{ m}^2$

$= 17\text{ m}^2\ \ \text{(round up)}$

♦♦ *Mean mark part (b) 30%.*

$\text{Cost of carpet} = 17 \times 40$

$= \$680$

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22. Measurement, STD2 M7 2015 HSC 29c

i. By measurement,

$\text{Roof length} = 1.5\text{ times the roof width}$

$\text{Width} = 12\text{ m}\ \ \text{(given)}$

♦ *Mean mark 35%.*

$\text{Length} = 1.5 \times 12$

$= 18\text{ m}$

$\therefore\ \text{Area of roof} = 12 \times 18$

$= 216\text{ m}^2\ \ \ldots\text{as required}$

ii. Volume of water

$= Ah$

$= 216 \times 0.005\ \ \ (5\text{ mm} = 0.005\text{ m})$

$= 1.08\text{ m}^3$

♦♦♦ *Mean mark 14%.*

$\text{Volume of cylinder} = \pi r^2 h$

$r = \dfrac{3.6}{2} = 1.8\text{ m}$

Find h when $V = 1.08\text{ m}^3$,

$\pi \times 1.8^2 \times h = 1.08$

$\therefore h = \dfrac{1.08}{\pi \times 1.8^2}$

$= 0.1061\ldots\text{ m}$

$= 10.6\text{ cm}\ \ \text{(to 1 d.p.)}$

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23. Measurement, STD1 M5 2020 HSC 28

Consider ΔABC:

$$\text{Area} = \frac{1}{2} \times AB \times BC$$

$$20 = \frac{1}{2} \times 8 \times BC$$

$$\therefore\ BC = 5$$

Using Pythagoras in ΔABC:

$$AC = \sqrt{8^2 + 5^2} = \sqrt{89}$$

♦♦♦ *Mean mark 11%.*

Since $\Delta ABC \ ||| \ \Delta DEF$,

$$\frac{AC}{BC} = \frac{DF}{EF}$$

$$\frac{\sqrt{89}}{5} = \frac{DF}{4}$$

$$\therefore\ DF = \frac{4\sqrt{89}}{5}$$

$$= 7.547...$$

$$= 7.55 \text{ cm (to 2 d.p.)}$$

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24. Measurement, STD1 M7 2021 HSC 21

$$\text{Distance walked on map} = 2 \times (8+5) = 26 \text{ cm}$$

♦♦♦ *Mean mark 24%.*

$$\text{Actual distance} = 26 \times 3000$$

$$= 78\,000 \text{ cm}$$

$$= 780 \text{ m}$$

$$= 0.78 \text{ km}$$

$$12 \text{ minutes} = \frac{12}{60} = 0.2 \text{ hours}$$

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{0.78}{0.2}$$

$$= 3.9 \text{ km/hr}$$